

Terminology

Alphabet (Σ): The 'Character class'

↳ A set of symbols

$\Sigma = \{0, 1\}$: binary alphabet

$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7\}$: octal alphabet

$\Sigma = \{a, b\}$

↳ a, ab, ba, abb

String: Sentence or Word

↳ Sequence of symbols

↳ Finite in length

$s = abbac$ length of $s = |s|$

Empty String (ϵ)

↳ a string where $|\epsilon| = 0$

Language: A set of language strings over some fixed alphabet

Example:

$L_1 = \{a, baa, bccb\}$

$L_2 = \{\}$

$L_3 = \{\epsilon\}$

↳ They are different

$L_4 = \{\epsilon, ab, abab, ababab, \dots\}$

↑ each string is finite in length but set may contain infinite elements

Terminology

Prefix of string ^{removing}

↳ String obtained by [↑]zero or more trailing symbols

$S = \text{Hello}$

Prefixes: $\epsilon, H, He, Hel, Hell, Hello$

Suffix of string

↳ String obtained by ^{removing} [↑]zero or more leading symbols

$S = \text{Hello}$

Suffixes: $\text{Hello}, \text{Hell}, \text{Hel}, \text{He}, \text{H}, \epsilon$

Proper Suffix & Proper Prefix:

$$PP = S[0] \neq \epsilon$$

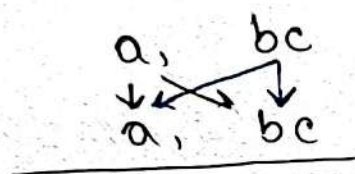
$$PS = S[n-1] \neq \epsilon$$

Subsequence:

A long sentence.

sub sequence

→ Kleene Closure



aa, abc, abc, bcb

$$\text{Let } L = \{a, bc\}$$

$$L^0 = \{\epsilon\}$$

$$L^1 = \{a, bc\}$$

$$L^2 = LL = \{aa, abc, bca, bcba\}$$

$$L^3 = LLL = \{aaa, aabc, abca, abcba, bcaa, bcabc, bcbaa, bcbcb\}$$

$$L^n = L^{n-1}L = LL^{n-1}$$

$\therefore$ Kleene Closure = 0 or more.

$$L^* = \left\{ \underbrace{\epsilon}_{L^0}, \underbrace{a, bc, aa, abc, bca, bcba}_{L^1}, \underbrace{\dots}_{L^2} \right\}$$

→ Positive Closure

1 or more

$$L^+ = L^* \text{ BUT } '\epsilon' \text{ not included}$$

Regular Expressions

$$a|b|cab = \{a, b, cab\}$$

$$b^* = \{\epsilon, b, bb, bbb, \dots\} \text{ // 0 or more}$$

$$a|b^* = \{a, \epsilon, b, bb, bbb, \dots\}$$

$$(a|b)^* = \{\epsilon, a, b, aa, ab, ba, bb, aaa, \dots\}$$

↳ set of all strings of a's and b's including ' ϵ '

$$L(a|b)^* = L(a^*) \cup L(b^*)$$

$$= \begin{array}{c} \epsilon, a, aa, aaa \\ \downarrow \swarrow \searrow \downarrow \swarrow \searrow \downarrow \swarrow \searrow \\ \epsilon, b, bb, bbb \end{array}$$

Algebraic laws of Regular Expressions

Failure Function

Q/The Fibonacci Strings

$$S_1 = b$$

$$S_2 = a$$

$$S_n = S_{n-1}S_{n-2} \quad n \geq 2$$

Construct failure function for S_6 .

$$S_3 = S_2 S_1 = ab$$

$$S_4 = S_3 S_2 = aba$$

$$S_5 = S_4 S_3 = abaab$$

$$S_6 = S_5 S_4 = \underbrace{abaab}_{S_5} \underbrace{aba}_{S_4}$$

$$\therefore S_6 = \boxed{abaababa}$$

String size	Proper Prefix	Proper Suffix
$ S =1$ $S=a$	—	—
$ S =2$ $S=ab$	a	b
$ S =3$ $S=aba$	$\boxed{a} = 1$ ab	$\boxed{a} = 1$ ba
$ S =4$ $S=abaa$	$\boxed{a} = 1$ ab aba a	$\boxed{a} = 1$ aa baa
$ S =5$ $S=abaab$	$\boxed{ab} = 2$ a aba abaa	$\boxed{ab} = 2$ b aab baab
$ S =6$ $S=abaaba$	$\boxed{aba} = 3$ a ab abaa abaab	$\boxed{aba} = 3$ a ba aaba baaba

Shayan

$$|s| = 7$$

$s = \text{abaabab}$

a
 $\boxed{ab}$
 aba
 abaa
 abaab
 abaaba

abab
 aabab
 baabab

$$|s| = 8$$

$s = \text{abaababa}$

a
 ab
 $\boxed{aba}$
 abaa
 abaab
 abaaba
 abaabab

a
 ba
 $\cdot \boxed{aba}$
 baba
 ababa
 aababa
 baababa

s	1	2	3	4	5	6	7	8
f(s)	0	0	1	1	2	3	2	3

↗ Context-Free Grammar

A grammar / formal system used to describe a class of languages,

Purpose of CFG:

- To list all strings in a language using a set of rules (production rules)
- Extends the capabilities of Regular Expressions (RE) & Finite Automata.

~~ab~~

A CFG is defined by—

1) Non-terminal Symbols / Variables :

↳ Abstract Category (e.g. $E, A, \dots$)

2) Terminal Symbols / Alphabet :

↳ Actual characters / Tokens,

↳ $id, +, -, a, b, \dots$

3) Production Rules :

Specify how non-terminals can be replaced with other non-terminals / terminals,

↳ $E \rightarrow E + E,$
 $E \rightarrow id \dots$

4) Starting Symbol :

The starting non-terminal,
which the grammar is first defined.